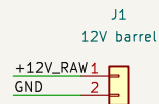


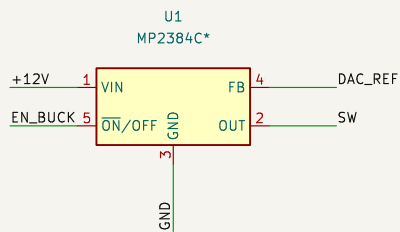
PocketForge Test-Node -- (d) POWER STAGE + FAIL-SAFE [infra-201 SECTION J]

CANDIDATE topology for the Fri EE review. Stub-routed net-label connectivity. * = Friday-gated value/MPN. [OSHW gear -> PCB silkscreen]

1. 12V IN + REV-POL P-FET

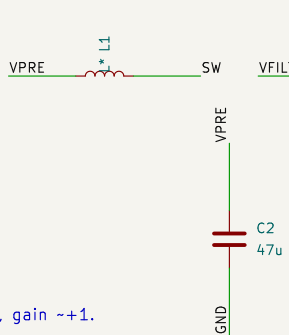


2. FPWM SYNC BUCK (12V->Vcell, forced-CCM)

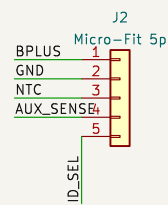
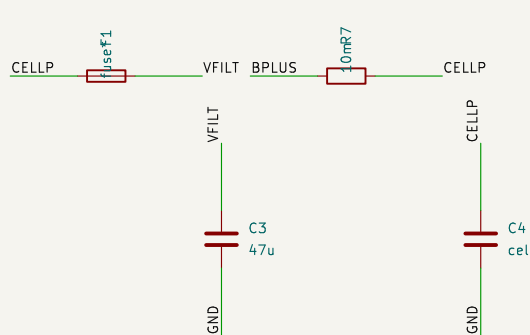


DAC_REF = MCP4728 ch-A (I2C) -> buck REFERENCE pin, gain ~+1.
(NOT FB-divider injection: a published design measured ~5.65x DAC-noise gain.)
EN_BUCK = ESP32 GPIO, default-OFF (LOW) at POR until a safe DAC code loads.

3. POST-FILTER (<1mV ripple)

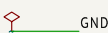
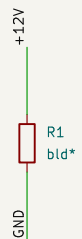
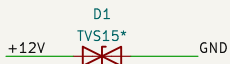


4. FUSE -> CELL NODE -> Micro-Fit

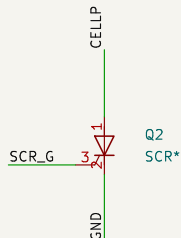
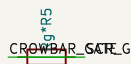
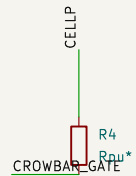
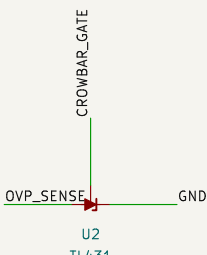


INA226 (bidir V+, after filter) -> ESP32 I2C slow DC-trim.
Kelvin sense: BPLUS + AUX_SENSE = remote-sense pair to the cell.

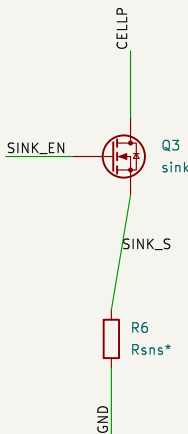
12V SHUNT (Input): bulk C / TVS / bleeder



5. INDEPENDENT OVP CROWBAR (>4.6V -> crowbar -> blow F1)



6. SINK-CLAMP (primary ~0.25A, also OVP assist)



CROWBAR: CELLP > ~4.6V -> TL431 (its OWN 2.5V ref, INDEPENDENT of the buck/DAC ref) sinks CROWBAR_GATE -> fires Q2 SCR -> shorts CELLP -> current through F1 blows it (~0.6 ms) -> isolates the device. D2 (TVS) catches the fast edge + is the crowbar-failed backstop.
CROWBAR-1 [RESOLVED, SPICE]: shorted buck HS-FET slams 12V at CELLP during sink. Worst-case CELLP peak = 4.7~5.0V (crowbar) / 6.5V (6-only, crowbar-failed) -- BOTH < ~7V AXP2202(=AXP717) BAT abs-max. The old 4.5V limit was the ADC/operating ceiling, NOT abs-max.
CROWBAR-2 [RESOLVED, SPICE]: peak is INSENSITIVE to crowbar latency 0.5~10us (147uF node + Lf di/dt set it). SCR ~96~99A pk, ~0.1J, ~0.6 ms conduction -> any small SCR handles it; size F1 I2t < SCR I2t. Friday: confirm with chosen MPNs + a bench HS-short pulse test.

SINK Q3 absorbs the AXP2202 ~254mA charge-back at low Vcell (sec-min dwell). SINK_EN = ESP32 GPIO.

POR (Friday): R1 12V-bleeder real W? D1 must clamp the 12V rail (a ~15~20V TVS, NOT a 5V part). Q3 shunt-vs-servo, armed-vs-gated.

GROUNDING (MAJOR -> grounding-ref schematic BEFORE KiCad layout): single-point STAR ground at C4; INA226 Kelvin pair to the star;

DAC ref + buck FB star-grounded by dedicated traces; pigtail B-/AUX_SENSE a dedicated/shielded pair.

STAGING: board #1 = pluggable rail -- populate BOTH this (d) switching rail AND a linear golden-reference rail (shared downstream)

to characterize AXP2202 gauge-ripple tolerance before committing the post-filter + 40 PCBs.